

WoW: A Window-to-Window Incremental Index for Range-Filtering Approximate Nearest Neighbor Search

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Introduction & Motivation

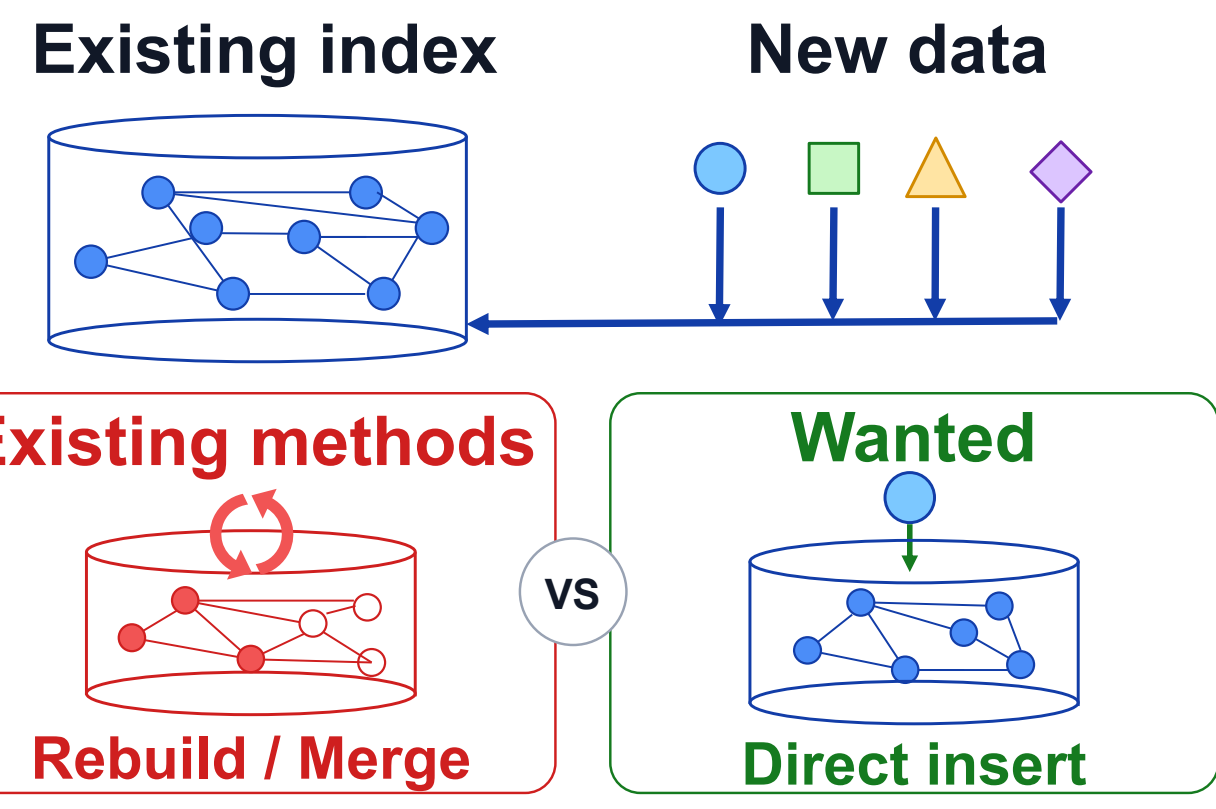
RFANNS: retrieve the k nearest vectors whose attribute values fall in a range filter, critical for e-commerce and RAG systems.

Database (Image + Price)



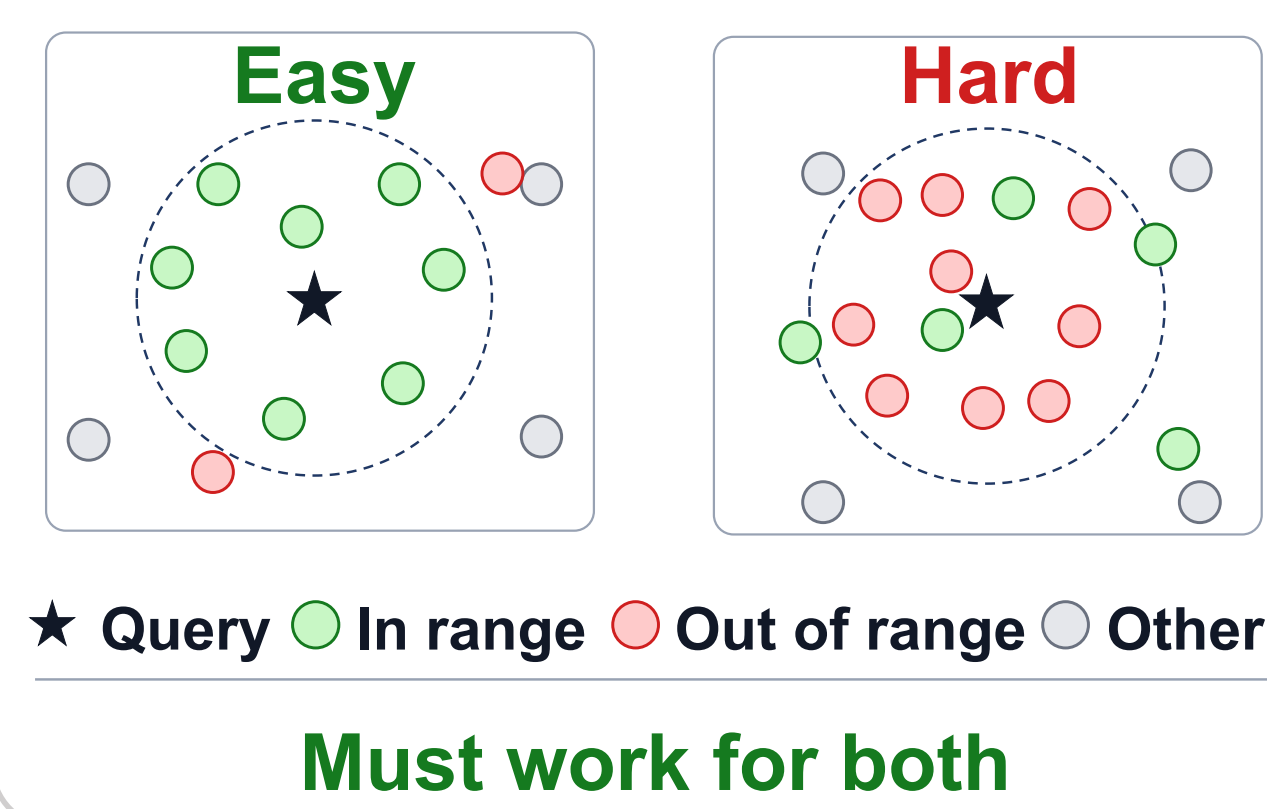
Challenge 1

Incremental update

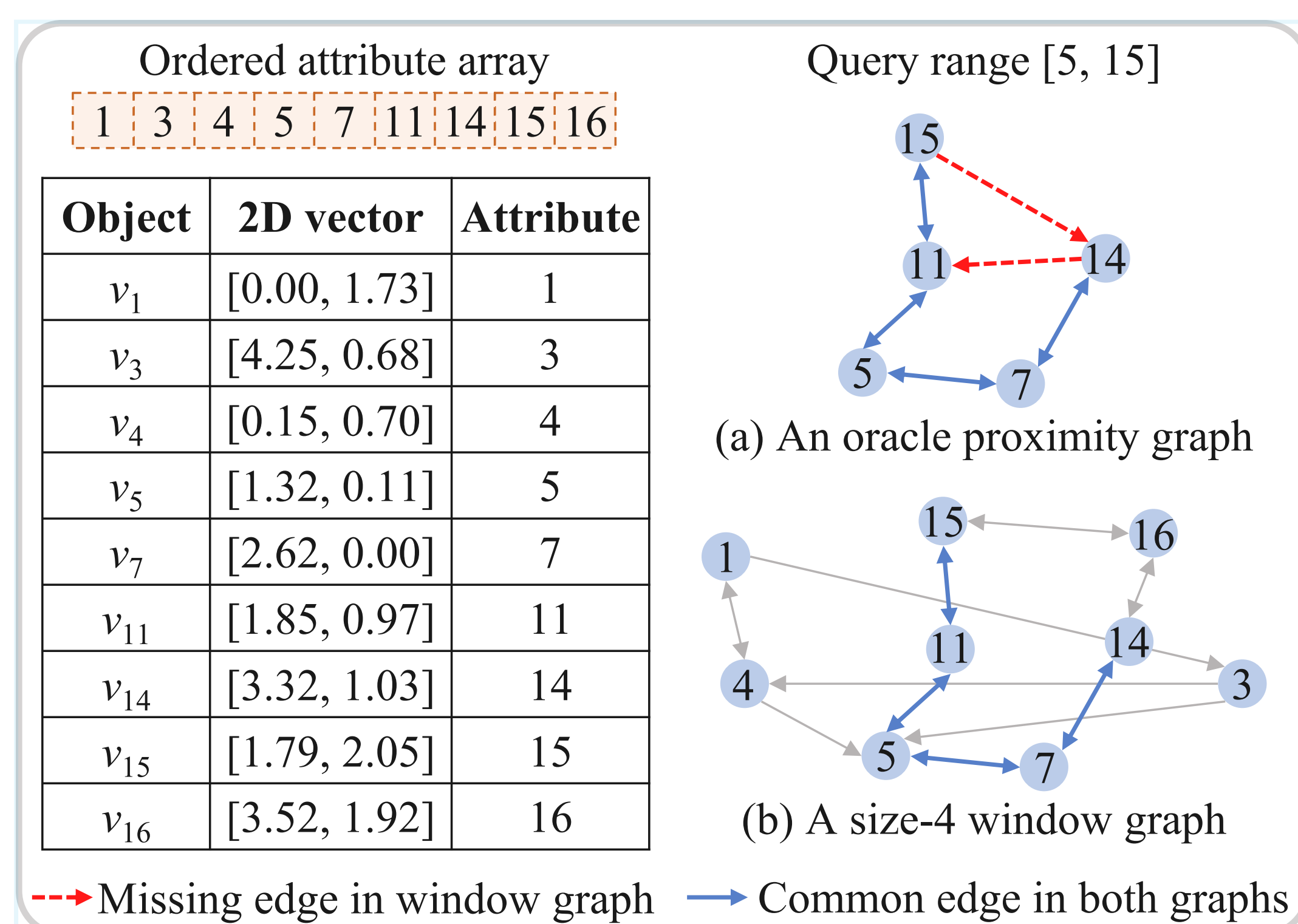


Challenge 2

Varying filters



Core Intuition: Window Graphs

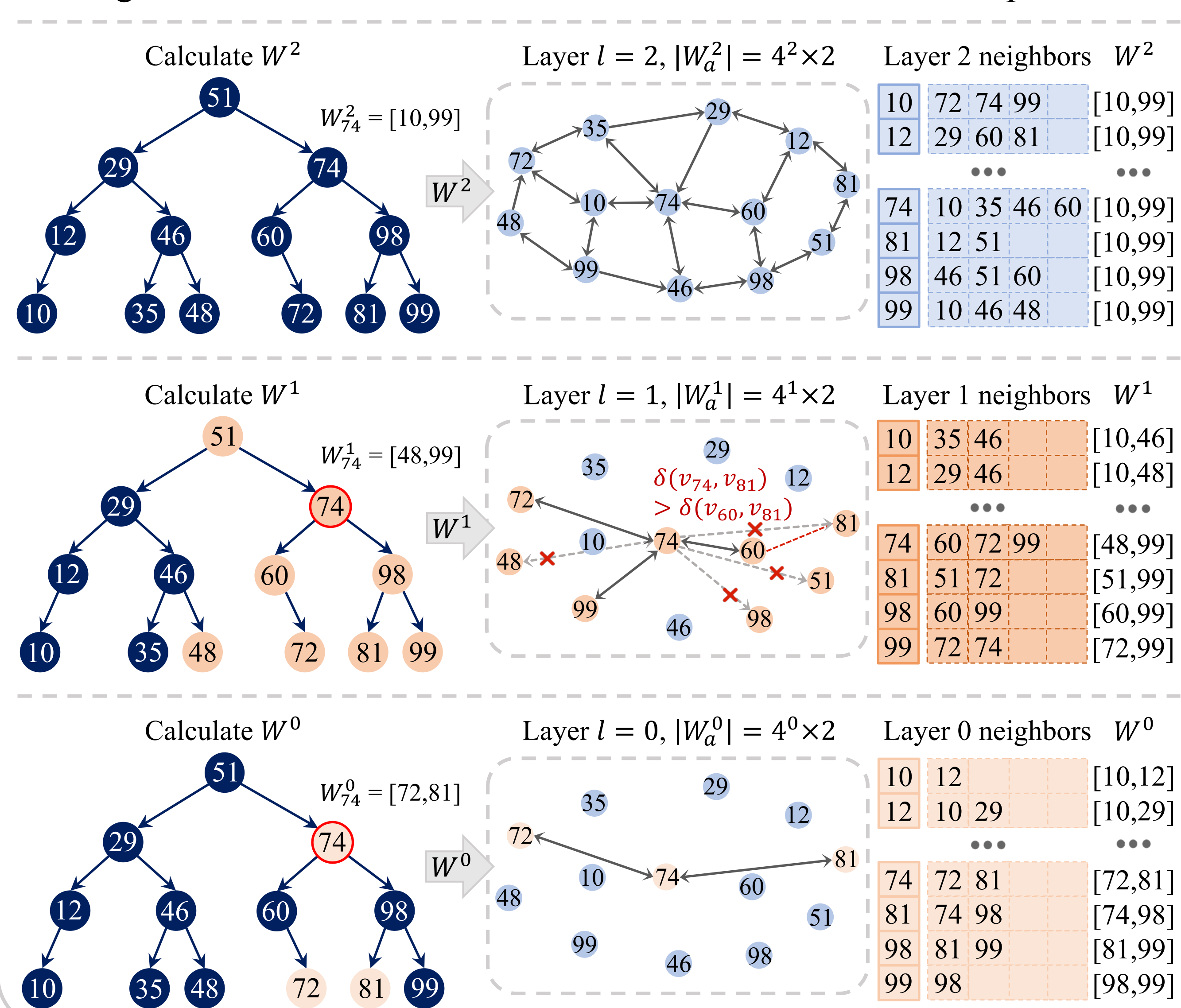


Window graphs connect vector and attribute similarity. (a) Oracle graph for range [5, 15]. (b) Size-4 window graph. Missing edges are recovered by smaller window graphs in the hierarchy.

WoW Index Structure

Weighted Balanced Tree

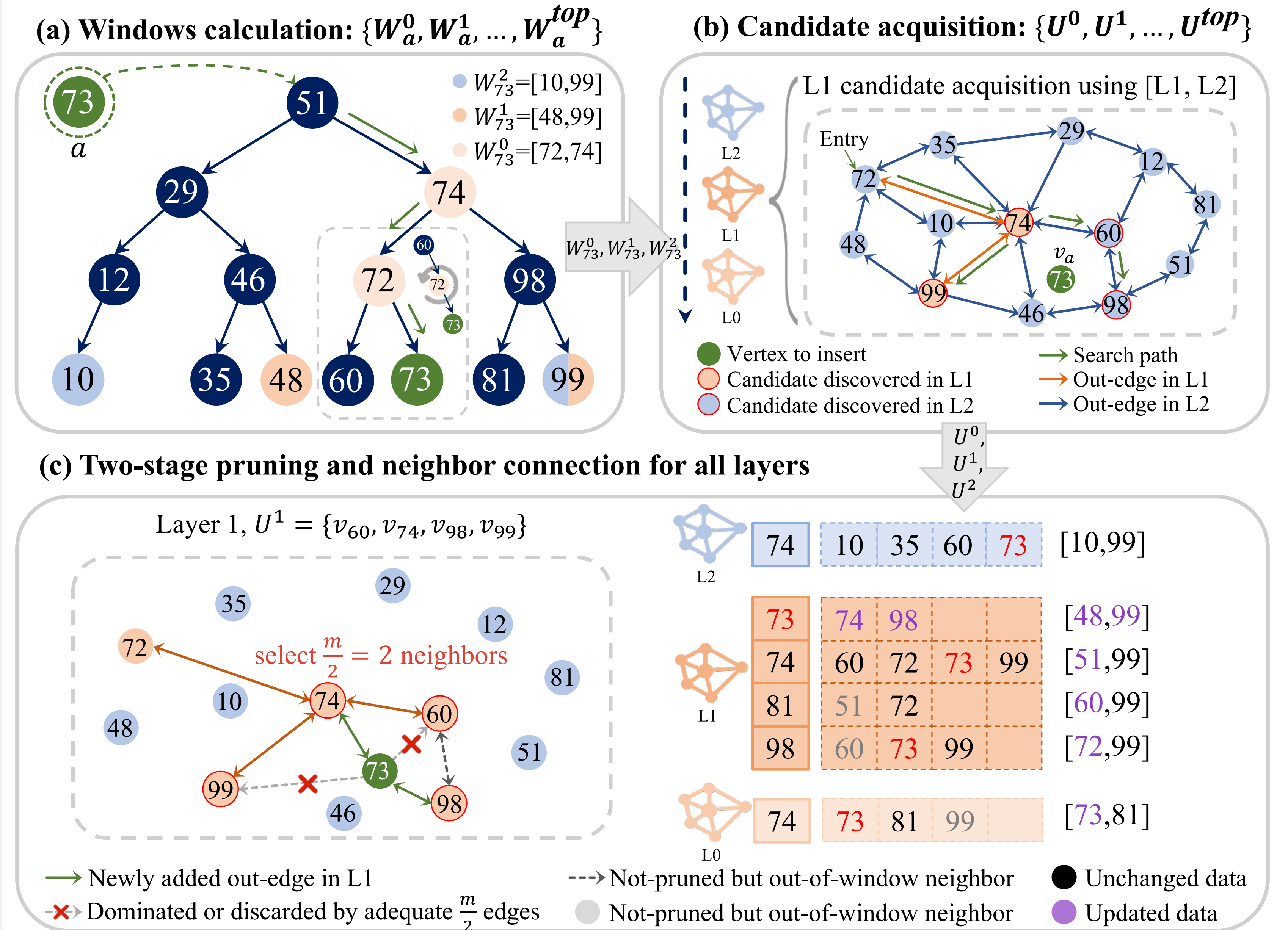
Hierarchical Window Graphs



WBT (left) + Hierarchical Window Graphs (right). WBT maintains windows in $O(\log n)$. $\mathcal{H} = \bigcup_{l \in [0, \text{top}]} G_l$ where each G_l has half-window size o^l . Edges satisfy both RNG and window properties.

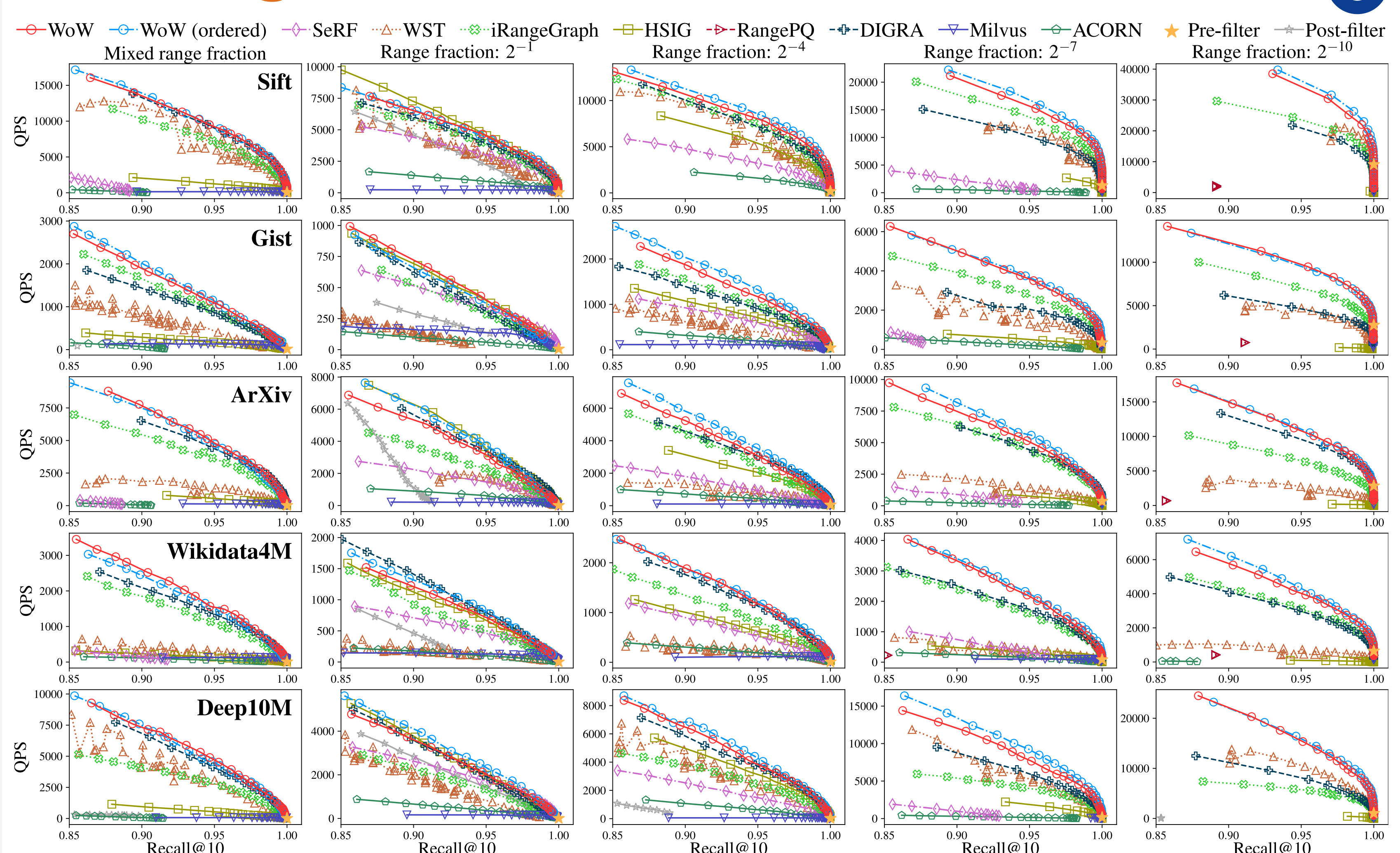
Insertion & Search

- Top-Down Insertion.** Compute windows via WBT; inherit candidates from upper layers (closer to target) before two-stage pruning. $O(\log^2 n)$ **worst-case**.
- Selectivity-Aware Query.** WBT computes in-range count n' to skip higher layers; early-stop skips lower layers when all neighbors are in range. $O(\log n')$ **query time**.



Insertion of v_{73} : (a) WBT computes windows; (b) beam search acquires candidates; (c) RNG pruning + two-stage neighbor adjustment.

QPS-Recall@10 — Main Results



5 datasets x 5 query selectivity workloads. From 2^{-10} (extreme) to 2^0 (low).

4x faster than the best increm. index and 1.5x faster than the best static index.

Index Size & Build Time

Index	Sift	Gist	ArXiv	Wiki4M	Deep10M
SeRF (size MB)	430	457	1,133	2,385	5,486
DIGRA (size MB)	1,533	1,533	3,547	9,501	21,477
WoW (size MB)	713	713	1,664	3,112	8,430
SeRF (time s)	112	591	592	3,053	1,574
DIGRA (time s)	1,935	11,236	15,230	49,178	51,312
WoW (time s)	152	391	623	1,557	3,360

4.9x faster than DIGRA with 0.4-0.5x smaller index size. 5-7x parallel speedup.

Conclusion & Links

- First** fully incremental RFANNS index with zero OOR distance computations.
- Selectivity-aware routing + WBT: **4.9x faster** indexing, **4x faster** queries.
- Scales to **100M vectors**, **5-7x** parallelism; $O(\log^2 n)$ insertion, $O(\log n')$ query.

Code: github.com/nju-websoft/WoW

Paper: PACMMOD Vol. 3, No. 6 (378)